Najm POC
Voice-to-Web Accident Report

Operational handoff, system flow, administration guide, and delivery roadmap

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Editable source: repository docs/*.md. This PDF is a generated release artifact.

1. Purpose and system flow

Najm is a proof of concept that receives a Hamsa voice-agent outcome, mints two private report links, and lets each accident party complete its own web section. It is demonstration-ready, not approved for real customer data or production operation.

End-to-end lifecycle

Driver calls Hamsa -> Hamsa sends lifecycle webhooks -> Najm records the live feed -> call.ended maps outcomeResult -> report and two opaque links are created -> SMS adapter sends the links -> parties complete their own web sections -> submissions update status, flags, and audit trail.

Key rules

- Every valid-looking webhook event is recorded; only call.ended mints a report.

- A repeated call.ended with the same callId returns the original report and links.

- Party A supplies shared accident details; both parties supply their own driver and vehicle details.

- Injury and assistive AI signals escalate a report; the app never decides fault automatically.

Operational surfaces

Surface	Use
/phone	Live webhook feed and simulated SMS phone frames. Open before a call.
/r/<slug>	Driver-facing flow, reached via opaque per-party link.
/dashboard/<reportId>	Report status, flags, parties, AI analysis, and audit trail.
localhost:4040	ngrok inspector for inbound webhook troubleshooting.

2. Run locally with ngrok and Hamsa

Prerequisites: Node 20+, npm, ngrok CLI/account, and Hamsa agent configuration access.

Start

1. npm install
2. cp .env.example .env.local
3. Set HAMSA_WEBHOOK_SECRET to a long random value.
4. npm run dev
5. Confirm http://localhost:3000/phone loads.

Expose

In terminal two run ngrok http 3000. Copy its HTTPS forwarding address and open https://<tunnel>/phone. Real webhook requests mint public links from this incoming host. NAJM_BASE_URL is primarily needed when the simulator must target the public tunnel.

Configure Hamsa

Setting	Value
URL	https://<tunnel>/api/webhook/hamsa
Method / type	POST / application/json
Authorization	Bearer followed by the same HAMSA_WEBHOOK_SECRET
Events	call.ended at minimum; lifecycle/transcription events for the live feed

Smoke test

POST a call.started event with the Bearer header to the public webhook. Expect {ok:true, recorded:call.started} and confirm the event appears in /phone. Inspect localhost:4040 first if it does not.

Outcome contract

Hamsa should return structured outcomeResult values on call.ended. The minimum useful capture is Party A name, mobile, plate, declared role, injury answer, and Party B mobile. Detailed synonyms and a sample payload are in docs/hamsa.md.

3. Administration and safety

Use synthetic data for demonstrations. Raw webhook envelopes are retained in the live feed for debugging, so this POC is not a production data store.

Area	Administrator action
Secrets	Keep .env.local private. Use a managed secret store in deployed environments and rotate at app and provider together.
Tunnel	Set a webhook secret before exposing ngrok. Update Hamsa whenever the free tunnel URL changes.
SMS	SMS failure does not block report creation. Check provider status and country/sender rules before retrying.
Data	Stop Next before deleting the local SQLite files for a clean demo. Never commit data copies or real transcripts.
Changes	Review code, Markdown source, and generated PDF together in one pull request.

First actions during an incident

- Secret exposed: stop external access, rotate it in the app and provider, restart/redeploy, then review scope without spreading PII.

- Unexpected traffic: stop ngrok, inspect tunnel/feed, and rotate the webhook secret if exposure is uncertain.

- Bad mapping: preserve a redacted example, update mapper and docs together, and add a regression fixture.

The complete operator guide is docs/admin-operations.md.

4. Working-demo checklist

This is the immediate operational work needed to demonstrate the existing POC end to end with Hamsa.

Task	Done when
D1 - D4: local setup	Node/npm/ngrok are available; .env.local has HAMSA_WEBHOOK_SECRET; npm run dev and npm run simulate:call work; /phone shows events and links.
D5: public tunnel	ngrok http 3000 supplies an HTTPS URL and its /phone page loads.
D6 - D8: Hamsa setup	Hamsa POSTs to /api/webhook/hamsa with the matching Bearer secret, call.ended enabled, and the required outcomeResult fields.
D9: smoke test	A public call.started request returns 200 and appears in /phone.
D10: real call	A completed Hamsa call shows link_minted and Party A/Party B SMS entries.
D11: complete flow	Both generated links submit and the dashboard reaches complete or expected escalation.
D12 - D13: rehearse	Decide simulated vs real Twilio SMS, rehearse fully, then reset local demo data.

Minimum Hamsa fields

full_name, mobile, plate_number, declared_role, injuries, and other_party_mobile. If fields do not prefill, inspect ignoredKeys and compare the actual payload with docs/hamsa.md.

5. Productionization roadmap

Do not proceed to real customer use until the P0 work below is complete.

Priority	Required outcome
P0	Privacy/threat model, production auth, managed secrets, managed encrypted data, verified webhook/replay defenses, observability, real integrations, and CI test suite.
P1	Reliable SMS, supportable link lifecycle, Arabic/RTL accessibility QA, human review operation, retention automation, and deployment pipeline.
P2	Complex multi-party cases, governed photo analysis, operational reporting, and load/resilience testing.

Recommended first two weeks

- Name technical, product, security/privacy, and Hamsa-integration owners.

- Complete privacy/threat modeling and confirm Hamsa authentication first.

- Stand up managed secrets, database, and observability in staging.

- Use the simulator payload as the first automated webhook regression fixture.

The full backlog and acceptance criteria are in docs/task-list.md.

6. Documentation control

Markdown under docs/ is the editable master and Git history is the version record. This PDF is regenerated for stakeholders who need a portable, offline snapshot.

Source document	Purpose
docs/run-local-hamsa.md	Setup, ngrok, Hamsa, smoke test, demo, and troubleshooting.
docs/flow.md	Architecture, lifecycle diagrams, safety boundaries, and operational surfaces.
docs/admin-operations.md	Access, secrets, routine operations, incidents, and release checks.
docs/task-list.md	POC-to-production backlog and sequence.
docs/README.md	Catalog, ownership, review cadence, and update process.

Regenerate: npm run docs:pdf

Before merging, inspect output/pdf/najm-poc-handoff.pdf and run npm run typecheck. Commit the generated PDF with its source Markdown changes.